THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **A Square and a Cube** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 2	$72 = 2^3 \times 3^2$; multiply by the primes with odd power
2	**(c)** 3	$192 = 2^6 \times 3$
3	**(a)** 9	Only the ones digit 7 matters: 7^2 ends in 9
4	**(c)** 1.2	1.44 is the square of 1.2
5	**(a)** 0.3	$0.027 = 0.3^3$
6	**(d)** 2025	Numbers ending in 5: 4×5 then 25 gives 2025
7	**(d)** 2	$98 = 2 \times 7^2$; multiply by the primes with odd power
8	**(c)** 2	$250 = 2 \times 5^3$
9	**(b)** 4	Only the ones digit 8 matters: 8^2 ends in 4
10	**(c)** 1.5	2.25 is the square of 1.5

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$2916 = 2^2 \times 3^6$ (1); $\sqrt{2916} = 54$ (1).	1+1
12	$\frac{\sqrt[3]{27}}{\sqrt[3]{512}}$ (1) $= \frac{3}{8}$ (1).	1+1
13	$\frac{\sqrt{81}}{\sqrt{256}}$ (1) $= \frac{9}{16}$ (1).	1+1
14	$4624 = 2^4 \times 17^2$ (1); $\sqrt{4624} = 68$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$1872 = 2^4 \times 3^2 \times 13$ (1); multiplier 13 (1); product 24336, root 156 (1).	1+1+1
16	$12000 = 2^5 \times 3 \times 5^3$ (1); divisor 12 (1); quotient 1000, root 10 (1).	1+1+1
17	$196 = 2^2 \times 7^2$ (1); root of the whole number 14 (1); so $\sqrt{1.96} = 1.4$ (1).	1+1+1
	OR: $2646 = 2 \times 3^3 \times 7^2$ (1); multiplier 6 (1); product 15876, root 126 (1).	1+1+1
18	$10976 = 2^5 \times 7^3$ (1); divisor 4 (1); quotient 2744, root 14 (1).	1+1+1
	OR: $324 = 2^2 \times 3^4$ (1); root of the whole number 18 (1); so $\sqrt{3.24} = 1.8$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Next perfect square above 1500 is $39^2 = 1521$ (2); students needed $= 21$ (2); rows $= 39$ (1).	2+2+1
	OR: $\sqrt[3]{1728} = 12$ cm (2); TSA $= 6s^2 = 6 \times 12^2 = 864$ cm² (3).	2+3
20	Next perfect square above 2000 is $45^2 = 2025$ (2); students needed $= 25$ (2); rows $= 45$ (1).	2+2+1
	OR: $\sqrt[3]{2744} = 14$ cm (2); TSA $= 6s^2 = 6 \times 14^2 = 1176$ cm² (3).	2+3

Mistakes to avoid:

- Concluding that a number is a perfect square from its last digit alone: a units digit of 0, 1, 4, 5, 6 or 9 is only *necessary*, not sufficient.
- Pairing primes for a perfect cube: grouping them in **pairs** (square-root rule) instead of **triples**.
- Writing $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$ or $(a+b)^2 = a^2 + b^2$.
- Mixing up 2^3 with 2×3 or 3^2 with 3×2; $4^3 = 64$, not 12.
- Saying the cube root of -8 does not exist; it is -2 because $(-2)^3 = -8$.
- Multiplying or dividing by the wrong prime when making a number a perfect square or cube, instead of looking only at the primes with an incorrect exponent.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **A Square and a Cube** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 0.5	$0.125 = 0.5^3$
2	**(a)** 4225	Numbers ending in 5: 6×7 then 25 gives 4225
3	**(a)** 2	$128 = 2^7$; multiply by the primes with odd power
4	**(c)** 11	$297 = 3^3 \times 11$
5	**(c)** 1	Only the ones digit 9 matters: 9^2 ends in 1
6	**(b)** 0.13	0.0169 is the square of 0.13
7	**(c)** 1.1	$1.331 = 1.1^3$
8	**(c)** 5625	Numbers ending in 5: 7×8 then 25 gives 5625
9	**(b)** 2	$162 = 2 \times 3^4$; multiply by the primes with odd power
10	**(d)** 3	$375 = 3 \times 5^3$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{\sqrt[3]{125}}{\sqrt[3]{216}}$ (1) $= \frac{5}{6}$ (1).	1+1
12	$\frac{\sqrt{169}}{\sqrt{441}}$ (1) $= \frac{13}{21}$ (1).	1+1
13	$6084 = 2^2 \times 3^2 \times 13^2$ (1); $\sqrt{6084} = 78$ (1).	1+1
14	$\frac{\sqrt[3]{343}}{\sqrt[3]{729}}$ (1) $= \frac{7}{9}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$3042 = 2 \times 3^2 \times 13^2$ (1); multiplier 2 (1); product 6084, root 78 (1).	1+1+1
16	$6750 = 2 \times 3^3 \times 5^3$ (1); divisor 2 (1); quotient 3375, root 15 (1).	1+1+1
17	$576 = 2^6 \times 3^2$ (1); root of the whole number 24 (1); so $\sqrt{5.76} = 2.4$ (1).	1+1+1
	OR: $4500 = 2^2 \times 3^2 \times 5^3$ (1); multiplier 5 (1); product 22500, root 150 (1).	1+1+1
18	$8624 = 2^4 \times 7^2 \times 11$ (1); divisor 1078 (1); quotient 8, root 2 (1).	1+1+1
	OR: $784 = 2^4 \times 7^2$ (1); root of the whole number 28 (1); so $\sqrt{7.84} = 2.8$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Next perfect square above 2500 is $51^2 = 2601$ (2); students needed $= 101$ (2); rows $= 51$ (1).	2+2+1
	OR: $\sqrt[3]{3375} = 15$ cm (2); TSA $= 6s^2 = 6 \times 15^2 = 1350$ cm² (3).	2+3
20	Next perfect square above 3000 is $55^2 = 3025$ (2); students needed $= 25$ (2); rows $= 55$ (1).	2+2+1
	OR: $\sqrt[3]{5832} = 18$ cm (2); TSA $= 6s^2 = 6 \times 18^2 = 1944$ cm² (3).	2+3

Mistakes to avoid:

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- Pairing primes for a perfect cube: grouping them in **pairs** (square-root rule) instead of **triples**.
- Writing $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$ or $(a+b)^2 = a^2 + b^2$.
- Mixing up 2^3 with 2×3 or 3^2 with 3×2; $4^3 = 64$, not 12.
- Saying the cube root of -8 does not exist; it is -2 because $(-2)^3 = -8$.
- Multiplying or dividing by the wrong prime when making a number a perfect square or cube, instead of looking only at the primes with an incorrect exponent.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **A Square and a Cube** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 9	Only the ones digit 3 matters: 3^2 ends in 9
2	**(d)** 2.5	6.25 is the square of 2.5
3	**(a)** 0.4	$0.064 = 0.4^3$
4	**(a)** 7225	Numbers ending in 5: 8×9 then 25 gives 7225
5	**(b)** 2	$200 = 2^3 \times 5^2$; multiply by the primes with odd power
6	**(b)** 2	$686 = 2 \times 7^3$
7	**(d)** 4	Only the ones digit 8 matters: 8^2 ends in 4
8	**(b)** 0.09	0.0081 is the square of 0.09
9	**(c)** 2	$242 = 2 \times 11^2$; multiply by the primes with odd power
10	**(b)** 50	$1350 = 2 \times 3^3 \times 5^2$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{\sqrt{225}}{\sqrt{784}}$ (1) $= \frac{15}{28}$ (1).	1+1
12	$8100 = 2^2 \times 3^4 \times 5^2$ (1); $\sqrt{8100} = 90$ (1).	1+1
13	$\frac{\sqrt[3]{64}}{\sqrt[3]{1331}}$ (1) $= \frac{4}{11}$ (1).	1+1
14	$\frac{\sqrt{289}}{\sqrt{576}}$ (1) $= \frac{17}{24}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$5292 = 2^2 \times 3^3 \times 7^2$ (1); multiplier 3 (1); product 15876, root 126 (1).	1+1+1
16	$10125 = 3^4 \times 5^3$ (1); divisor 3 (1); quotient 3375, root 15 (1).	1+1+1
17	$961 = 31^2$ (1); root of the whole number 31 (1); so $\sqrt{9.61} = 3.1$ (1).	1+1+1
	OR: $6760 = 2^3 \times 5 \times 13^2$ (1); multiplier 10 (1); product 67600, root 260 (1).	1+1+1
18	$9072 = 2^4 \times 3^4 \times 7$ (1); divisor 42 (1); quotient 216, root 6 (1).	1+1+1
	OR: $1296 = 2^4 \times 3^4$ (1); root of the whole number 36 (1); so $\sqrt{12.96} = 3.6$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Next perfect square above 3545 is $60^2 = 3600$ (2); students needed $= 55$ (2); rows $= 60$ (1).	2+2+1
	OR: $\sqrt[3]{9261} = 21$ cm (2); TSA $= 6s^2 = 6 \times 21^2 = 2646$ cm² (3).	2+3
20	Next perfect square above 4100 is $65^2 = 4225$ (2); students needed $= 125$ (2); rows $= 65$ (1).	2+2+1
	OR: $\sqrt[3]{15625} = 25$ cm (2); TSA $= 6s^2 = 6 \times 25^2 = 3750$ cm² (3).	2+3

Mistakes to avoid:

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